



Man on Mars: 2030

NASA might launch its manned expedition to Mars if all goes right <http://dgit.in/Mars2030>



Human Genome decoding

A set of recently unveiled machines by Illumina can process 18,000 human genome samples in a year. Read more here: <http://dgit.in/HumGeno>

Space Age



Mondays must really suck in this neighborhood of the Universe.

The Largest Black Hole

As we know black holes are formed when stars collapse and begin to condense in size but not mass, forcing a concentrated area of highly compressed matter which exerts huge gravitational pull in the region, so much so that even light is said to be consumed by it, thus giving it the name black hole. The paradox of black holes is that the smaller they may be in size the greater they are in mass. As the most massive cosmic phenomena, black holes impact not only their surrounding regions but also the balance of all things due to their immense gravity.

The largest supermassive black hole in the observable universe is the quasar OJ287 that is calculated to be 18 billion times the mass of our sun and is found in the centre of its galaxy in the constellation Cancer. To truly imagine its size a cosmic scale would be necessary, in the case of OJ287 it would be an object that is capable of eclipsing our very galaxy in term of size.

A possible successor to this blackhole is the yet unconfirmed supermassive black hole in the centre of the NGC1277 galaxy in the constellation Perseus. This blackhole is believed to be 17 billion times the mass of the sun and is composed of nearly 14 per cent of this galaxy's mass. In perspective, the event horizon of this blackhole is eleven times the distance of Neptune from the Sun or roughly 300 AU (diameter of the solar system is 61 AU).

Largest Man Made Structures in Space

In the face of the Universe's immense scale the largest space structure made

by humanity seem like nothing but specks of dust. However their very ambition is a testament to the human potential – not only in their success but in their symbolism, which embraces unity of nations and scientific cooperation.

The International Space Station is the largest man made structure ever to be launched and can even be seen orbiting the Earth in the clear night sky. The ISS was assembled in modules with the first steps in November 1998. Since then over 150 components, 1,000 plus hours of spacewalks and a steady roster of dedicated astronauts, cosmonauts and scientists have kept building this structure despite setbacks. The total mass of the station is over 400,000 kilograms and is expected to reach 400 tonnes when it is finally completed by April 2014.

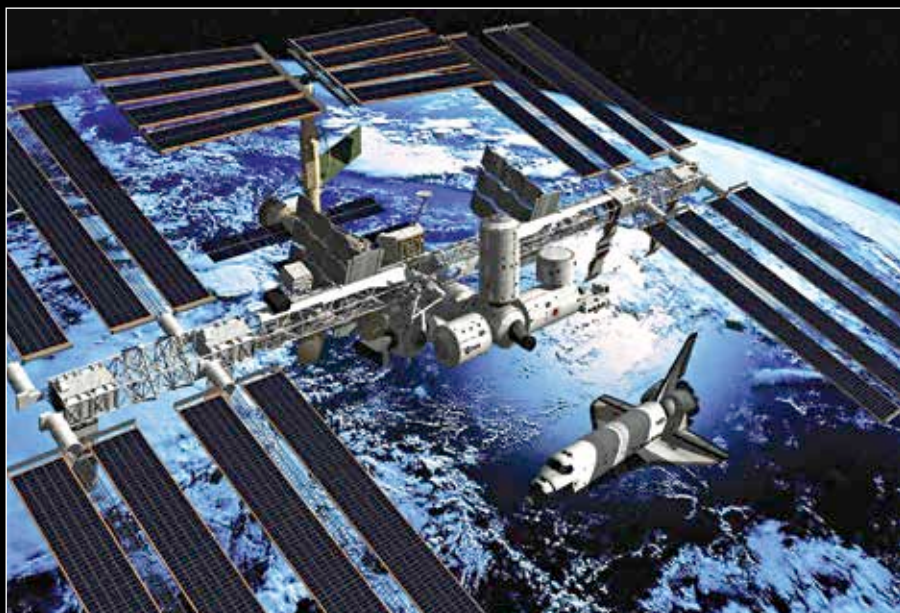
The ISS is an example of massive large scale cooperation between various nations that include the United States, Russia, Japan and those in Europe, who collectively

a symbol for the commitment of humanity towards its future that will surely involve adventures towards space.

Humanity Dreams Biggest

Carl Sagan said that the visions we offer future generations shape the future and dreams are the maps to discovering them. The many astounding undertakings by humanity such as the ISS show how at times even conflicted groups of nations can collaborate towards a common peaceful goal. This ability to set aside our differences for what truly matters is the true largeness of the human spirit - one that can't be contained by just one planet.

Its worthwhile to remember that as the Voyager 1 space probe was leaving the Solar System in 1990, after having finished its primary mission, Carl Sagan made a small request of NASA - that it be rotated so it could take and transmit an image of how the Earth looked from 6 billion kilometers away. This image came



The ISS has been occupied for over 4825 days and has orbited the Earth nearly 87,000 times, and will continue till 2020.

embarked on this scientific and education mission. The 73 meter long station orbits the Earth 15.7 times per day with a steady rotating international crew from 15 different nation that has inhabited the station for over 13 years. By setting aside historic disagreements and overcoming international conflicts, the ISS stands as

to known by its iconic name - the Pale Blue Dot - forever becoming a reminder of the unimaginable scale of the universe and evidence of humanity's humble existence in it. It is important that even in the face of that humbling realisation humanity has continued to follow its better angles and look upwards to the sky with awe. d